[1] Professor Rachid Yazami's passion for science started when he was a child in his native city, Fez, with a fascination for geology. His academic ambitions soon progressed from analyzing rocks to acquiring a doctorate in graphite compounds for lithium batteries in France. In 1980, aged 26, he made a key discovery that led to the development of the lithium-graphite anode, which is used in rechargeable batteries in mobile phones and laptops.

[2] "I still remember the day I opened my battery and saw that graphite had become a gold color. I converted graphite into gold," he told Rebecca from Inspire Middle East. His professors were very supportive when they realized the importance of his discovery: lithium can be stored into graphite. This was a turning point in battery history.

[3] Despite this scientific achievement, it took more than a decade to commercialize the lithium battery. For some reason, the French government didn't consider the invention of graphite as a major invention. So, Yazami did not get a copyright, which means that **it** could be used for free by other companies. Today, around 95% of batteries produced globally use Yazami's technology. Worldwide, analysts think that the commercial lithium battery market is worth about $80 billion.

[4] In his battery research, Yazami is exploring the capabilities of artificial intelligence for ameliorating the quality of people's lives. He is working on a technique that will allow an electric car battery to last for up to ten years instead of five years now. Moreover, the battery can be charged in 10 minutes, the same time we spend in the gas station when we fill our car tank.

[5] Yazami has won many awards, including the Draper Prize, the equivalent of a Nobel Prize for engineers. Even at the age of 67, he remains dynamic and wholly dedicated to his work. He is currently the Principal Scientist at Nanyang Technological University in Singapore, and is frequently consulted by the business community **there**.

Adapted from: www.euronews.com

I. COMPREHENSION (15 POINTS)

BASE ALL YOUR ANSWERS ON THE TEXT.

A. ARE THESE STATEMENTS TRUE OR FALSE? JUSTIFY. (3 pts)

1. Professor Yazami's first interest was lithium batteries.

. . . . . .

. . . . . .

2. The majority of batteries worldwide use Yazami's technology.

. . . . . .

. . . . . .

3. Yazami has an important position at Nanyang Technological University in Singapore.

. . . . . .

. . . . . .

B. ANSWER THESE QUESTIONS. (3 pts)

1. How has Professor Yazami's invention changed the history of batteries?

. . . . . .

2. How long did it take the lithium batteries to be on the market?

. . . . . .

3. Why could companies use Yazami's invention for free?

. . . . . .

C. COMPLETE THE FOLLOWING SENTENCES WITH INFORMATION FROM THE TEXT.(2pts)

1. Thanks to Yazami's technique, electric car batteries will . . . . . .

. . . . . .

2. Yazami is still active although . . . . . .

. . . . . .

D. PICK OUT FROM THE TEXT PHRASES OR SENTENCES WHICH SHOW THAT (2 pts)

1. Yazami was encouraged by his professors.

. . . . . .

. . . . . .

2. the Draper Prize is a very important prize.

. . . . . .

. . . . . .

E. FIND IN THE TEXT WORDS THAT MEAN ALMOST THE SAME AS (3 pts)

1. very important (paragraph 1): . . . . . .

2. improving (paragraph 4): . . . . . .

3. committed (paragraph 5) : . . . . . .

F. WHAT DO THE UNDERLINED WORDS IN THE TEXT REFER TO? (2 pts)

1. it: (paragraph 3): . . . . . .

2. there: (paragraph 5): . . . . . .

II. LANGUAGE (15 POINTS)

A. PUT THE WORDS IN BRACKETS IN THE CORRECT FORM. (2 pts)

1. John needed a special medical (treat) . . . . . after the accident.

2. Although the story is simple, it teaches important (morality) . . . . . lessons.

B. FILL IN THE GAPS WITH THE RIGHT WORDS FROM THE LIST. (2 pts)

launch - cultural - strange - foreign - achieve

1. Our university will . . . . . a new training program for its staff.

2. The new economic reform will attract . . . . . investment in agriculture.

C. REWRITE THE SENTENCES AS INDICATED. (4 pts)

1. The book was cheap. Yet it did not sell well.

The book . . . . . in spite of . . . . . .

2. They will clean the room after the party.

The room . . . . . .

3. “Did Amina call the doctor?” The neighbor asked.

The neighbor wanted to know . . . . . .

4. The journal rejected Amal’s article because she used impolite language.

The journal . . . . . if she hadn’t used impolite language.

D. PUT THE VERBS IN BRACKETS IN THE CORRECT FORM. (3 pts)

1. My car (break) . . . . . down three times last week.

2. We started painting the house yesterday. We (paint) . . . . . the living room and the kitchen by tomorrow afternoon.

3. Many students are interested in (join) . . . . . our yoga club.

E. MATCH EACH EXPRESSION WITH ITS APPROPRIATE FUNCTION. (2 pts)

Expressions	Functions
1. “How do you find the idea of working on Saturdays?” 2. “Would you mind helping me fix my washing machine?” 3. “I’ll understand better if you give me an example.” 4. “You should get your parents’ permission before you travel alone.”	a. Making a complaint b. Asking for clarification c. Giving advice d. Making a request e. Asking for opinion

1. …….. 2. …… 3. …… 4. ……

F. COMPLETE THE FOLLOWING EXCHANGES APPROPRIATELY. (2 pts)

1. Rachid: . . . . . .

Lisa: Great news! Let’s celebrate it!

2. Karim:. . . . . .

Hind: Good idea! The weather is warm in Essaouira these days.

DO TASKS 1 AND 2

TASK 1: Complete the following paragraph (4 pts)

There are many things I would like my school administration to change.

First, ...

TASK 2: (6 pts)

Do you prefer to travel with your friends or with your family?

Write a short article to be published in a travel blog about your choice. Give your reasons.

KEY AND MARKING SCALE

Accept any appropriate answer not mentioned in this key.

I. COMPREHENSION (15 POINTS)

Testees should not be penalised for spelling or grammar mistakes in the comprehension section.

TRUE / FALSE: (3 pts)

1. **False.** "His passion for science started… with a fascination for geology." (paragraph 1)
2. **True.** "Around 95% of batteries produced globally use Yazami's technology." (paragraph 3)
3. **True.** "He is currently the Principal Scientist at Nanyang Technological University in Singapore." (paragraph 5)

WH QUESTIONS: (3 pts)

1. His discovery that lithium can be stored into graphite was a turning point in battery history; it led to the lithium-graphite anode used in rechargeable batteries.
2. It took more than a decade.
3. Because Yazami did not get a copyright (the French government did not consider the invention as a major one).

SENTENCE COMPLETION: (2 pts)

1. …last for up to ten years instead of five (and be charged in 10 minutes).
2. …he is 67 years old.

PICK OUT: (2 pts)

1. "His professors were very supportive."
2. "The equivalent of a Nobel Prize for engineers."

RIGHT WORD MEANING: (3 pts) 1. key 2. ameliorating 3. dedicated

WORD REFERENCE: (2 pts) 1. (Yazami's) invention 2. Singapore

II. LANGUAGE (15 POINTS)

WORD FORMATION: (2 pts) treatment · moral

GAP FILLING: (2 pts) launch · foreign

REWRITING SENTENCES: (4 pts)

1. The book **did not sell well** in spite of **being cheap**.
2. The room **will be cleaned after the party**.
3. The neighbor wanted to know **if Amina had called the doctor**.
4. The journal **wouldn't have rejected Amal's article** if she hadn't used impolite language.

VERB FORMS: (3 pts) broke · will have painted · joining

MATCHING: (2 pts) 1. e. Asking for opinion 2. d. Making a request 3. b. Asking for clarification 4. c. Giving advice

EXCHANGES: (2 pts)

1. I've just passed my driving test! / I got the highest mark in the exam!
2. Why don't we spend the weekend in Essaouira? / How about going to Essaouira this weekend?

III. WRITING (10 POINTS; task 1: 4 pts; task 2: 6 pts)

The following criteria should be respected while scoring the writing tasks.

Scoring criteria	Task 1	Task 2
Relevance of content to the task	2 pts	3 pts
Appropriateness and variety of vocabulary	1 pt	2 pts
Grammar and mechanics	1 pt	1 pt

Model answers (any appropriate answer is accepted)

Task 1: There are many things I would like my school administration to change. First, I would like them to build a bigger library with computers, because the present one is too small. Second, they should repair the playground and buy sports equipment so that students can practise more activities. Moreover, I think they should organize more clubs, such as theatre and English clubs. Finally, the school should provide healthier food in the canteen. If these changes happen, our school will be a better place to learn.

Task 2: Travelling with Friends or Family? Many people wonder whether it is better to travel with friends or with family. Personally, I prefer to travel with my friends. First of all, friends share the same interests, so we can choose activities that we all enjoy, such as hiking or visiting museums. Moreover, travelling with friends is more fun because we laugh a lot and make unforgettable memories. Another reason is that we have more freedom: we can wake up late, change our plans and try new things without asking permission. In addition, sharing the costs makes the trip cheaper. However, I also enjoy trips with my family from time to time. In conclusion, travelling with friends is the best choice for me because it is exciting and full of freedom.